

Julian Vara

909-800-0013 | jvara6282@gmail.com | linkedin.com/in/julian-vara/ | github.com/jvara34 | jvara34.github.io

EDUCATION

California State University of San Bernardino

Bachelor of Science in Computer Science, Minor in Data Science GPA: 3.88

Aug. 2024 – Present

San Bernardino, CA

Riverside City College

Associate of Science in Math and Science GPA: 3.30

Aug. 2022 – June 2024

Riverside, CA

EXPERIENCE

Student Research Assistant

Aug. 2025 – Present

California State University, San Bernardino

- Researching deep learning methods for cancer detection classification by experimenting with CNNs, ResNet, VGGNet, GoogLeNet, and EfficientNet. Utilized Python and Jupyter notebook for the extensive library in Deep Learning.
- Building and refining scalable neural network architectures on a High Performance Computing (HPC) cluster using TensorFlow and Keras to enhance training efficiency and predictive accuracy. Co-authoring a research-level paper with faculty detailing methods, results, and implications

Undergraduate Data Science Researcher

Aug. 2025 – Dec. 2025

The Data Mine, Purdue University

- Developing a GPU-accelerated license plate recognition system using Python, OpenCV, TensorFlow, and NVIDIA CUDA
- Optimizing image recognition models with V2X engineers to improve detection reliability across varied conditions while expanding expertise in computer vision, text classification, and high-performance computing. Delivered technical presentations to project stakeholders/shareholders

Student Research Assistant

June 2025 – August 2025

California State University of San Bernardino

San Bernardino, CA

- Reduced 17K+ predictors down to 20 without sacrificing accuracy by applying penalized regression with distance correlation, streamlining model complexity.
- Improved predictive performance by lowering MSE from 0.3255 to 0.2299 and optimizing AIC/BIC values through advanced regression modeling. Ran large-scale simulations and data analyses in R/RStudio.

Student Research Assistant

July 2024

California State University of San Bernardino

San Bernardino, CA

- Interpreted a dataset from NASA to develop a prediction model for missing data on the semi-major axis of exoplanets. Using R/Rstudio for data manipulation and creation of the log transformation model.
- Improved the model with a log transformation, raising the R-squared to 99% and reducing RMSE to 0.162.

PROJECTS

Stalk.AI | Python, React, HTML, FastAPI, MUI, Vercel

August 2025 – Dec. 2025

- Developed a stock prediction web platform analyzing the top 100 publicly traded companies, providing buy/sell recommendations and risk assessments using deep learning models
- Implemented Firebase Authentication with secure login/signup flows, admin user management, and role-based access control; designed Firestore schemas for users, predictions, and financial data.
- Integrated deep learning models into a FastAPI powered backend, supporting inference and data exchange within a full stack pipeline

SKILLS

Languages & Frameworks: Python, R/RStudio, C++, JavaScript, FastAPI, React, Type Script

Data Science & ML: Regression (Ridge/Lasso), Feature Selection, Model Evaluation, TensorFlow, Pandas, NumPy, CNNs, Computer Vision (ResNet, VGG, EfficientNet)

Tools: Git/GitHub, Firebase, Linux, LaTeX, Google Suite, Microsoft Office

Research & Communication: Technical Writing, Presentations, Stakeholder Communication